

Kinematics of the the Merger Remnant

SUHANI SURANA¹

¹*University of Arizona, Tucson*

Keywords: Local Group — Major Merger — Rotation Curve — Velocity Dispersion — Galaxy — Galaxy Evolution

1. INTRODUCTION

Local group consists of the Milky Way (MW) and Andromeda (M31). The Local Group’s three most massive members are all spiral galaxies: the Milky Way (MW), Andromeda (M31), and Triangulum (M33), with their masses in a ratio of $\sim 10:10:1$. Local Group, which contains more than 80 galaxies and has a total mass of roughly $3.5 \times 10^{12} M_{\odot}$ (R. E. González et al. (2014)). They are on a collision course and will have their closest approach in about 4.3 Gyrs and a series of close approaches later. The future evolution of the Local Group is primarily shaped by the major merger between MW and M31. Studing the kinematics of such merger remnants will help us to placing the Local Group in the broader context of galaxy formation and evolution. A major merger refers to the gravitational collision and subsequent combination of two galaxies of roughly equal size and mass. The study of these major mergers can help us to understand the place of the Local Group in the cosmic evolution and can help us to further our understanding of the properties of cold dark matter.

A **galaxy** is a gravitationally bound set of stars whose properties cannot be explained by a combination of baryons (gas, dust and stars) and Newton’s laws of gravity (B. Willman & J. Strader (2012)). **Galaxy evolution** is the process by which galaxies change over time. This includes internal changes like star formation, aging stars, and gas depletion, as well as external events such as mergers and collisions. These processes affect a galaxy’s color, brightness, size, shape, and structure, leading to features like tidal tails, bridges, and changes in the interstellar medium. By investigating the stellar kinematics in simulated MW/M31 merger remnants, we can better constrain the physical mechanisms responsible for shaping the morphological and dynamical outcomes of major mergers like MW-M31. Also, the MW-M31 merger is expected to result in a transition from two well-structured spiral galaxies to a more spheroidal or elliptical remnant (A. Toomre & J. Toomre (1972)).

However recent studies have showed that disc galaxies could also be created from a major merger. Studying the resulting stellar kinematics—particularly of the bulge and disk components—provides valuable insight into the structural evolution of such galaxies. Also, investigating the kinematics of the stellar disk and bulge in the merger remnant tells us how angular momentum is redistributed and whether disk-like structures can reform after such a major merger event.

The merger between MW and M31 has a closest approach of $31.0^{+38.0}_{-19.8}$ kpc (R. P. van der Marel et al. (2012)). The resulting merger remnant will likely resemble an elliptical galaxy. M33 is expected to settle into orbit around the remnant, although there are non-negligible probabilities of a direct hit with the MW or even ejection from the Local Group (R. P. van der Marel et al. (2012)). The figure above illustrates the kinematic signatures of merger remnants, showing how properties such as ellipticity, and rotation vary with time. This supports findings from simulations where dissipational (gas-rich) mergers produce more oblate, rotating, and isotropic remnants, matching observed features of low-luminosity elliptical galaxies (T. J. Cox et al. 2006). Mergers (and in particular, major events) can result in S0 remnants with realistically coupled bulge-disc structures in less than ~ 3 Gyr (Querejeta, M. et al. (2015)). These remnants also lie on a fundamental plane (FP). Collisionless hierarchical mergers reproduce this FP with realistic scatter and tilt, the latter driven by increasing dark matter fraction in more massive remnants (D. Taranu et al. (2015)).

Despite considerable progress in understanding the MW–M31 merger, several important questions remain open. Simulations show that gas-rich mergers produce more rotationally supported, isotropic remnants, while dry mergers result in slowly rotating, anisotropic systems (T. J. Cox et al. (2006)). However, it is still unclear which specific combination of gas content, orbital geometry, and feedback processes will dominate in the actual

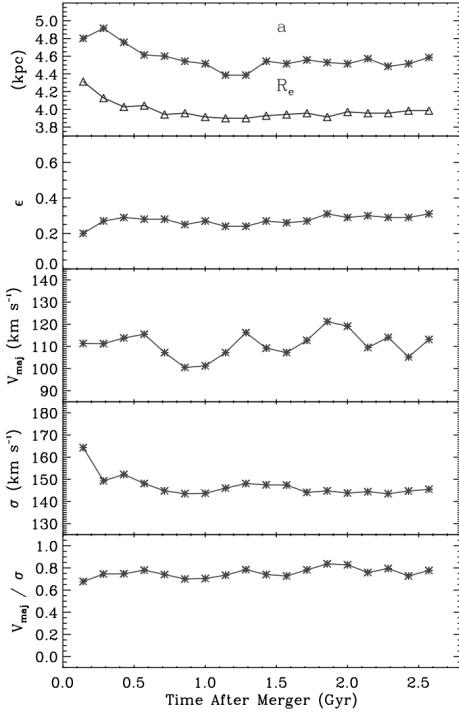


Figure 1. Time evolution of structural and kinematic properties of a major merger remnant: semimajor axis a , half-mass radius R_e , ellipticity ϵ , major axis rotation speed V_{maj} , velocity dispersion σ , and the ratio V_{maj}/σ . Based on simulations of identical disk galaxy mergers, relevant to MW–M31 remnant studies. Adapted from [T. J. Cox et al. \(2006\)](#). Settled dynamically within ~ 1 Gyr. Retained some disk-like features (moderate ellipticity, nonzero V/σ). V/σ stays between ~ 0.6 and ~ 0.75 , with a mild increasing trend. Thus, system is pressure supported (elliptical-like)

MW–M31 event. (([R. P. van der Marel et al. 2012](#))). Addressing these questions requires high-resolution simulations. My project specifically contributes to this effort by analyzing the expected kinematic profiles of the MW–M31 merger remnant using outputs from such simulations ([R. P. van der Marel et al. \(2012\)](#)), to better constrain its future rotational support and velocity dispersions.

2. THIS PROJECT

The topic of this project is Stellar disk/bulge kinematics after a major merger. I would be particularly looking at the evolution of the rotation curve. The rotation curve of a disc galaxy is a plot of the orbital speeds of visible stars or gas in that galaxy as a function of their radial distance from that galaxy’s centre ([Wikipedia contributors \(2025\)](#)) The “observed” rotation curve and the mass-derived rotation curve based on the enclosed mass profile would be compared which will offer insights into how well observational proxies

trace underlying mass distributions after a merger. Furthermore, the project aims to analyze the velocity dispersion components of the stellar populations in both the disk and bulge regions. This is essential for understanding whether the remnant will settle into a pressure-supported elliptical-like structure or retain significant rotational support.

The open question of whether the gas rich merger is more rotationally supported or is pressure supported will be understood by this project. We will also come to know the sense of rotation that this merger remnant exhibits post merger.

Understanding post-merger stellar kinematics is crucial to advancing our knowledge of galaxy evolution. The kinematics of merger remnants affect where they lie on the Tully–Fisher and Fundamental Plane. By comparing these results with observations, we can answer whether such remnants preserve rotational properties or rapidly settle into isotropic, pressure-supported systems which will tell us about fate of the MW–M31 remnant, Local group and the general formation knowledge of early-type galaxies.

3. METHODOLOGY

An N-body simulation refers to a computational technique in which the motion of a large number of particles is calculated by numerically solving their mutual gravitational interactions over time. The N-Body simulation used here is by [R. P. van der Marel et al. \(2012\)](#). They conducted collisionless N-body simulations of the MW–M31–M33 system, modeling the gravitational interactions between stars and dark matter components. The simulations were carried out using the GADGET-3 code. In the simulations, the dark halo of each galaxy was represented by Hernquist density profile ([L. Hernquist \(1990\)](#)).

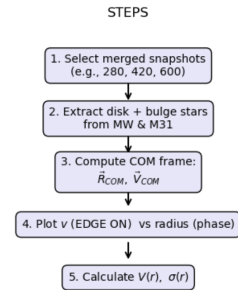


Figure 2. Steps to the project. The half mass radius method was adapted from [T. J. Cox et al. \(2006\)](#)

To study the kinematics of the merger remnant, I will first identify simulation snapshots where the MW and

M31 have fully merged and the system is dynamically relaxed. These snapshots will be by the separation plots in Homework 6. Using the formula $(\text{SN} * 10)/0.7$, I found out the snapshots numbers to be 280, 420 and 600. I will use high res as it would give me much better velocity dispersions as a function of r . Stellar particles belonging to the disk and bulge components of both the MW and M31 were extracted and combined into a single dataset representing the stellar body of the remnant. The center-of-mass (COM) position and velocity were computed using mass-weighted averages:

$$\mathbf{R}_{\text{COM}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i}, \quad \mathbf{V}_{\text{COM}} = \frac{\sum m_i \mathbf{v}_i}{\sum m_i}, \quad (1)$$

where m_i , $\mathbf{r}_i$, and $\mathbf{v}_i$ are the mass, position, and velocity of each stellar particle. All particle coordinates were shifted into the COM frame using:

$$\mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{COM}}, \quad \mathbf{v}'_i = \mathbf{v}_i - \mathbf{V}_{\text{COM}}. \quad (2)$$

To study rotational structure, the system was reoriented so that the net angular momentum vector lies along the z -axis, aligning the stellar disk (if present) in the xy -plane. A phase-space diagram plotting the velocity V against radius was constructed to visually identify rotation trends. This diagram was color-coded to show direction of motion and extended to 10–15 kpc to encompass any outer rotational features, consistent with methods from Lab 7.

Rotational support was quantified by computing both the “observed” rotation curve, defined by the mean V in radial bins, and the theoretical circular speed:

$$V_c(r) = \sqrt{\frac{GM(< r)}{r}}, \quad (3)$$

where G is the gravitational constant, $M(< r)$ is the mass enclosed within radius r , and r is the radius in kiloparsecs. This comparison reveals the extent to which the remnant is rotationally or pressure-supported.

The velocity dispersion profile was calculated in radial bins, especially within the half-mass radius, to trace how random motions vary with radius. The half-mass radius R_{half} is defined such that:

$$\sum_{i: R_i < R_{\text{half}}} m_i = \frac{1}{2} \sum_i m_i$$

To further classify the kinematic state, the ratio V/σ was determined, where V is the mean rotational velocity and σ the velocity dispersion. For each radial bin $[r_i, r_{i+1}]$, we compute the mean azimuthal velocity $V(r)$ and its dispersion $\sigma(r)$.

$$V(r) = \langle v_i \rangle_r, \quad \sigma(r) = \sqrt{\langle v_i^2 \rangle_r - \langle v_i \rangle_r^2}$$

. Then, the ratio V/σ will be plotted as a function of radius. Regions where $\max(\frac{V}{\sigma}) \gg 1$ indicate disk-like fast rotators, whereas values of $(\frac{V}{\sigma}) \ll 1$ correspond to pressure-supported ellipticals.

Plots of the rotation curves, dispersion profile, and V phase diagram, were plotted.

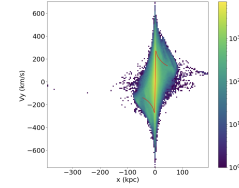


Figure 3. This is the observed velocity plotted as a function of radius and the circular velocity of the merger remnant edge on.

The MW/M31 merger remnant might exhibit rotational support. High velocity dispersion is expected due to the violent relaxation process which would be similar to elliptical galaxies.

4. RESULTS

Figure 5 shows the observed stellar velocity as a function of radius for the merger remnant at three different times: 4.00, 6.00, and 8.57 Gyr, alongside the expected circular velocity for an edge-on orientation. At 4.00 Gyrs, the V_y vs x plot is highly asymmetric. We can see two distinct streams which are MW and M31 colliding. The red curve of circular velocity (from the enclosed mass) does not tightly trace the dense points because many orbits are highly perturbed or they are radial. At 6 Gyrs, we can see a structure starting to form. A residual rotation signature remains so that part of the dense clump lies closer to the circular velocity line, indicates a rotation-supported component. By 8.57 Gyrs, the remnant looks like a single galaxy. The high-density region of stars is centered near y component of the velocity and spans both positive and negative velocities fairly evenly. The theoretical circular velocity traces our plot of V_y and x really tightly indicating that it is in a fairly stable circular orbit.

Figure 6 shows the evolution of the velocity dispersion as a function of radius over the course of the MW-M31 merger. At 4 Gyrs, the merger is still going on, so as we move towards different values of r , there is a strong variation which indicates that there are some regions that are dispersion supported while there are others which are rotation supported. At 6 Gyrs, the parts which are

merged show low velocity dispersion and the outer regions are rotationally supported. At 8.57 Gyrs, we see that the majority of the galaxy (within half mass radius) is dispersion supported. This also shows us that when the galaxies merge, the rotationally supported merger remnants change into dispersion supported merger remnants after the merger has happened.

5. CONCLUSION

The main takeaway from this paper is that after the merger has happened, the merger remnant is pressure supported. Yes, the result agrees with my hypothesis. In the first paragraph, we saw that the merger remnant after the merger event was stable and the theoretical circular velocity was well fitted into the V_y vs x plot. The MW- M31 merger is expected to result in a transition from two well-structured spiral galaxies to a more spheroidal or elliptical remnant (A. Toomre & J. Toomre (1972)). Thus, our result supports this expectation. Un-

derstanding this relation between the observed V_y vs x and theoretical circular velocity vs x tells us whether such remnants preserve rotational properties or rapidly settle into isotropic, pressure-supported systems which will tell us about fate of the MW-M31 remnant, Local group and the general formation knowledge of early-type galaxies. The theoretical circular velocity can be model dependent. Also, I used the half mass radius to filter out the v_{phi} . Also, I took the radius upto 45 kpc. It might not have included some points which could have made the results slightly different.

ACKNOWLEDGMENTS

I would like to thank Dr. Gurtina Besla for her invaluable guidance and inputs for the successful completion of the project. I would also like to extend my gratitude to the course TA, Himansh Rathore whose invaluable support and guidance have helped me to complete this project.

REFERENCES

- Cox, T. J., Dutta, S. N., Di Matteo, T., et al. 2006, The Astrophysical Journal, 650, 791, doi: [10.1086/507474](https://doi.org/10.1086/507474)
- González, R. E., Kravtsov, A. V., & Gnedin, N. Y. 2014, ApJ, 793, 91, doi: [10.1088/0004-637X/793/2/91](https://doi.org/10.1088/0004-637X/793/2/91)
- Hernquist, L. 1990, Astrophysical Journal; (USA), 356, doi: [10.1086/168845](https://doi.org/10.1086/168845)
- Querejeta, M., Eliche-Moral, M. C., Tapia, T., et al. 2015, AA, 579, L2, doi: [10.1051/0004-6361/201526354](https://doi.org/10.1051/0004-6361/201526354)
- Taranu, D., Dubinski, J., & Yee, H. K. C. 2015, The Astrophysical Journal, 803, 78, doi: [10.1088/0004-637X/803/2/78](https://doi.org/10.1088/0004-637X/803/2/78)
- Toomre, A., & Toomre, J. 1972, ApJ, 178, 623, doi: [10.1086/151823](https://doi.org/10.1086/151823)
- van der Marel, R. P., Besla, G., Cox, T. J., Sohn, S. T., & Anderson, J. 2012, The Astrophysical Journal, 753, 9, doi: [10.1088/0004-637X/753/1/9](https://doi.org/10.1088/0004-637X/753/1/9)
- Wikipedia contributors. 2025, https://en.wikipedia.org/w/index.php?title=Galaxy_rotation_curve&oldid=1284313750
- Willman, B., & Strader, J. 2012, The Astronomical Journal, 144, 76, doi: [10.1088/0004-6256/144/3/76](https://doi.org/10.1088/0004-6256/144/3/76)

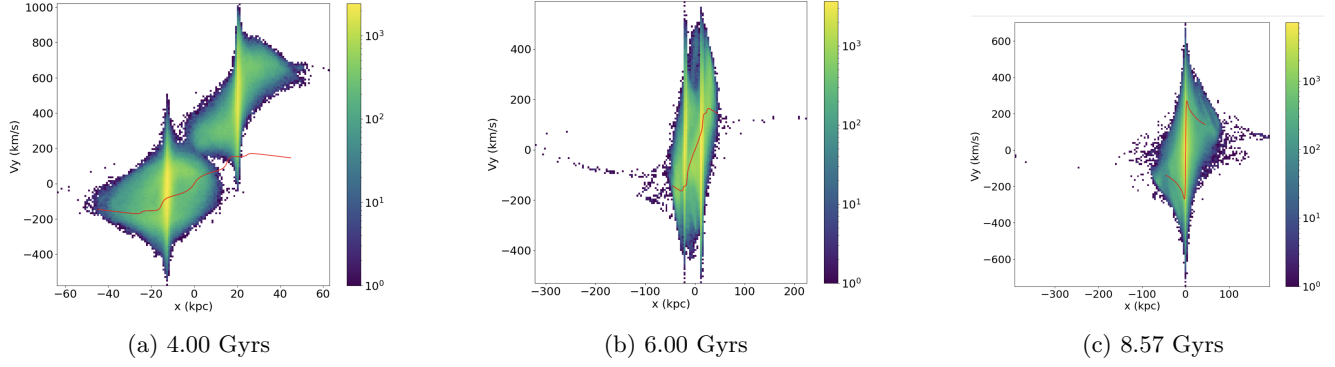


Figure 4. This is the observed velocity plotted as a function of radius in the merger remnant at 4.00, 6.00 and 8.57 Gyr (left to right) and the theoretical circular velocity of the merger remnant edge on. When the merger started, at 4 Gyrs the stars from both the galaxies form these clusters which have an opposite-sense motion, indicative of two merging cores or tidal streams. They also show a strong deviation from the theoretical circular velocity. At 6 Gyrs, most stars have merged into one system. There is still some deviation present from the theoretical circular velocity. At 8.57 Gyrs, it just seems like another galaxy with mostly stable orbits and the circular velocity follows the observed velocity. Thus, the galaxy at the beginning does not appear to be in stable circular orbit but after the merger, it seems to be in agreement with the calculated circular velocity.

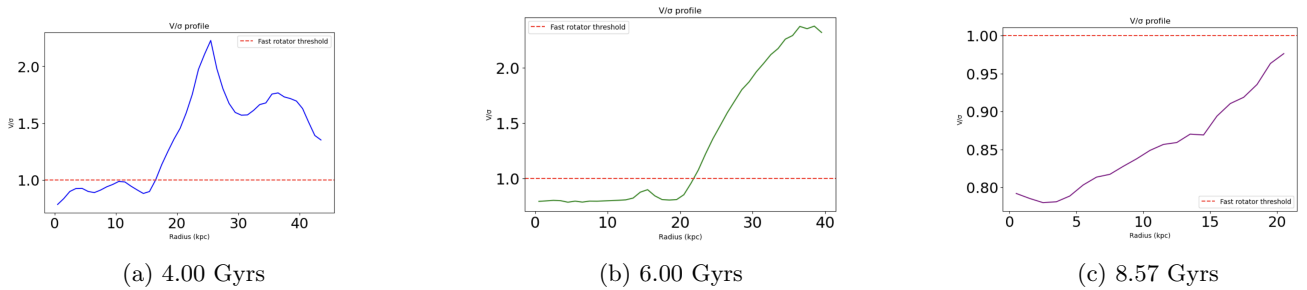


Figure 5. Evolution of the V/σ profile as a function of radius at three different stages of the galaxy merger: (a) 4.00 Gyrs, (b) 6.00 Gyrs, and (c) 8.57 Gyrs. The red dashed line indicates the threshold $V/\sigma = 1$, above which a galaxy is considered a fast rotator. At 4 Gyrs, the system shows peaks above the threshold in the outskirts, indicating local rotation-dominated regions due to merging substructures. By 6 Gyrs, a more coherent rotation pattern emerges at larger radii, but the inner regions remain dispersion-dominated. At 8.57 Gyrs, the system settles into a more stable configuration with low V/σ throughout, which hints at a transition to a dispersion-supported remnant.